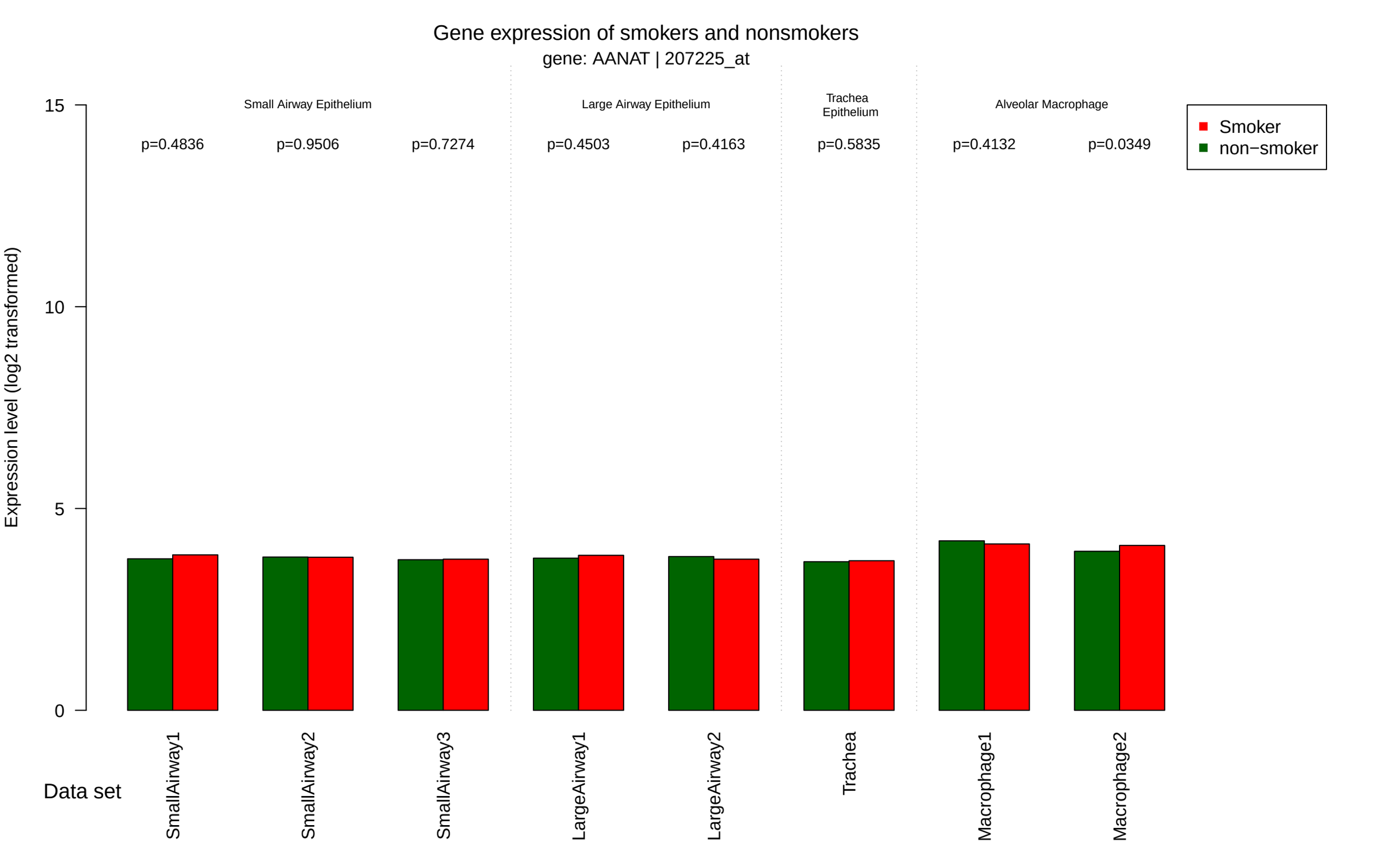
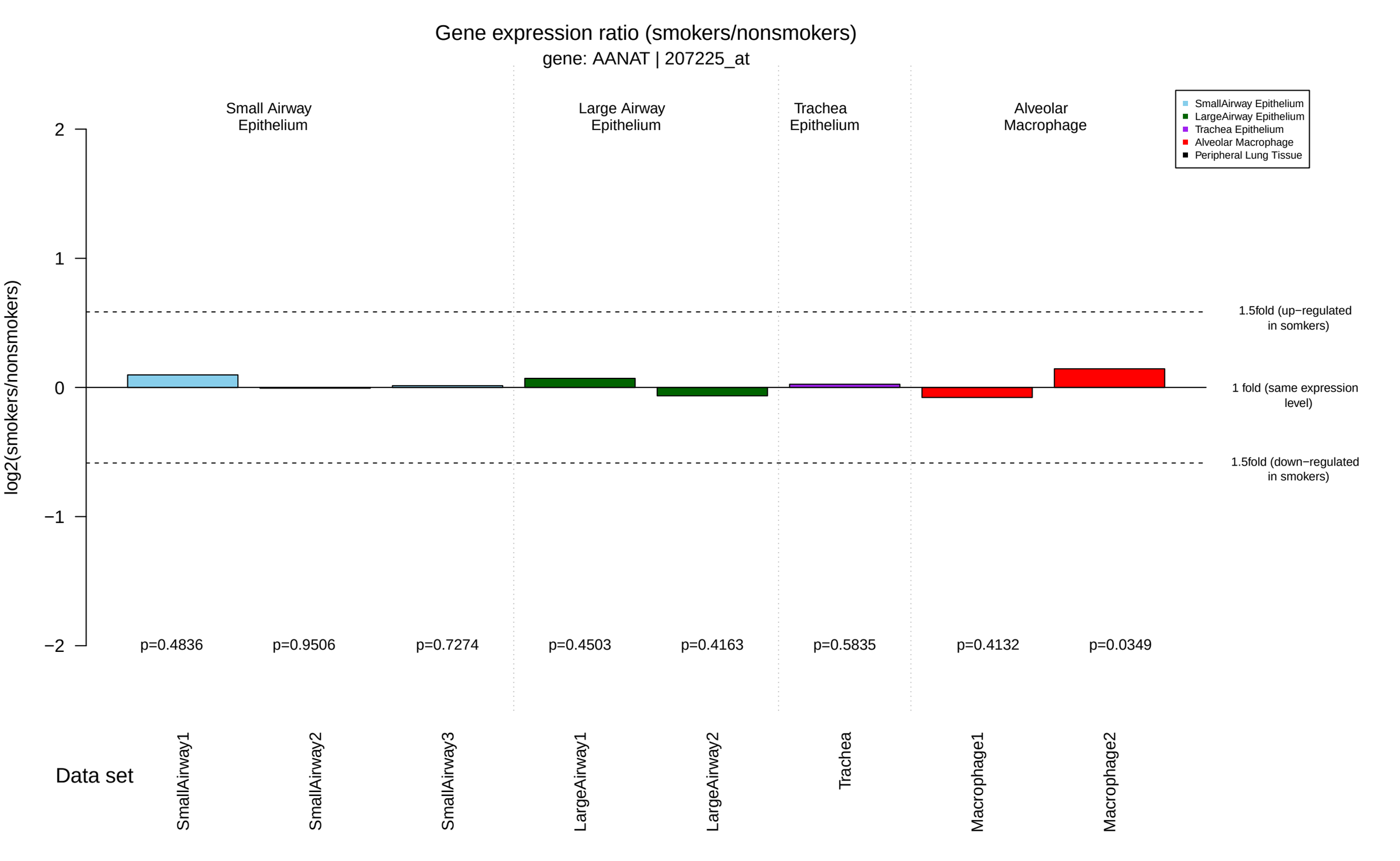
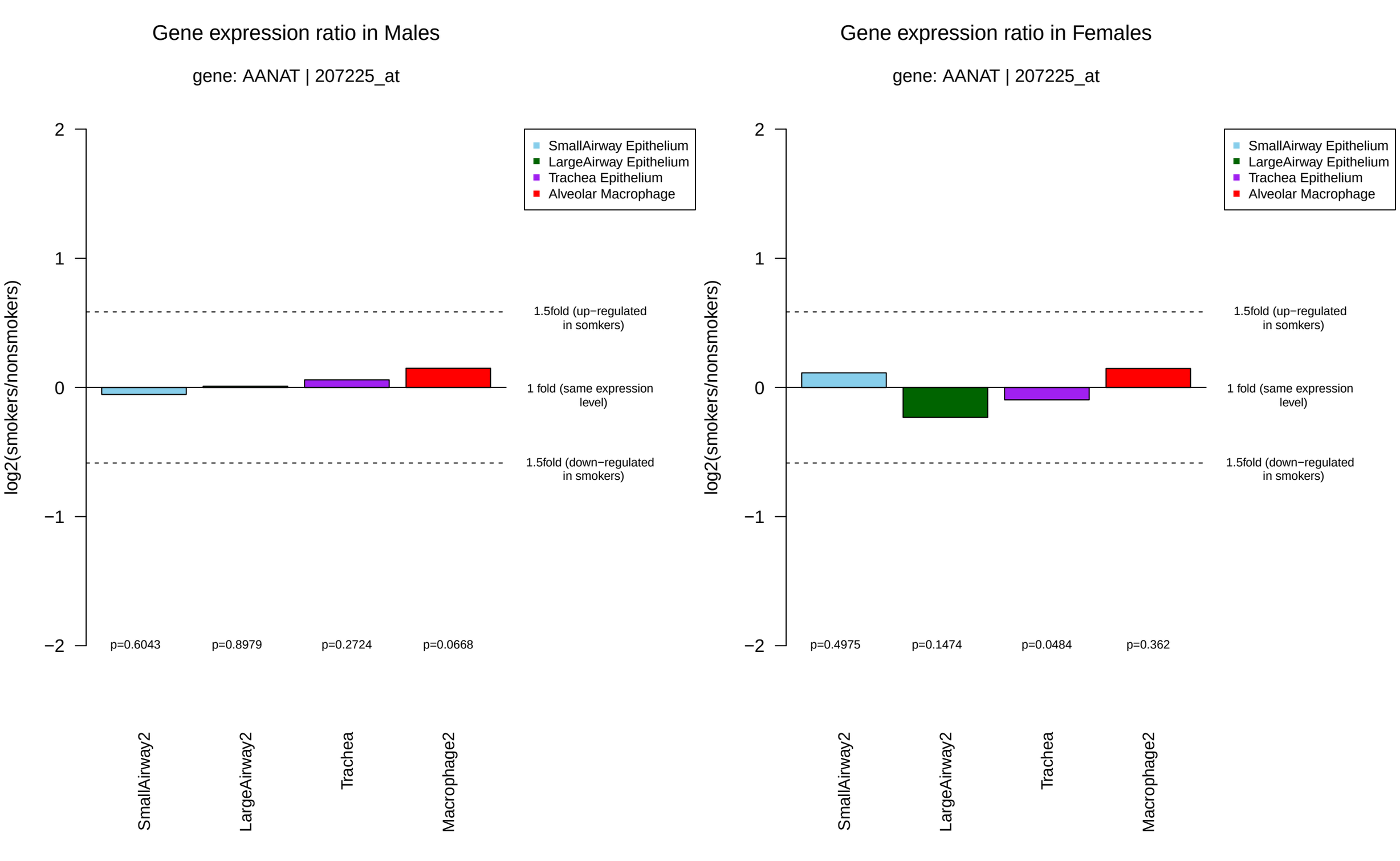
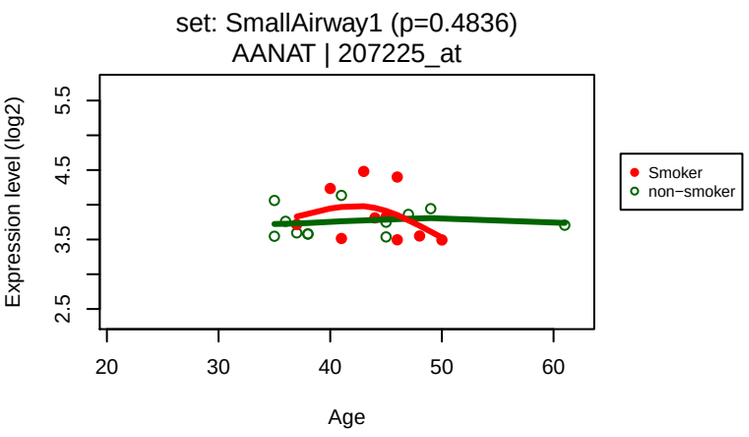
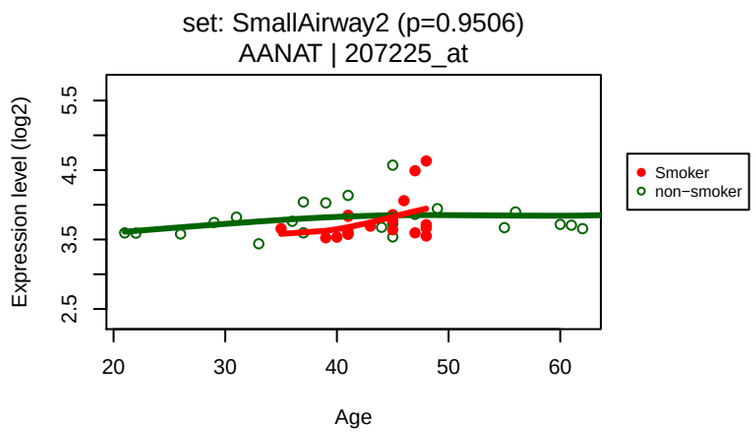
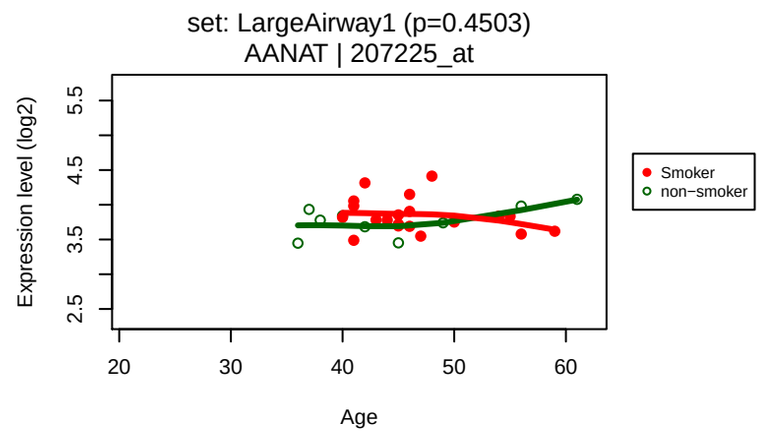

Gene expression of smokers and nonsmokers
gene: AANAT | 207225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4836
p=0.9506
p=0.7274
p=0.4503
p=0.4163
p=0.5835
p=0.4132
p=0.0349
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

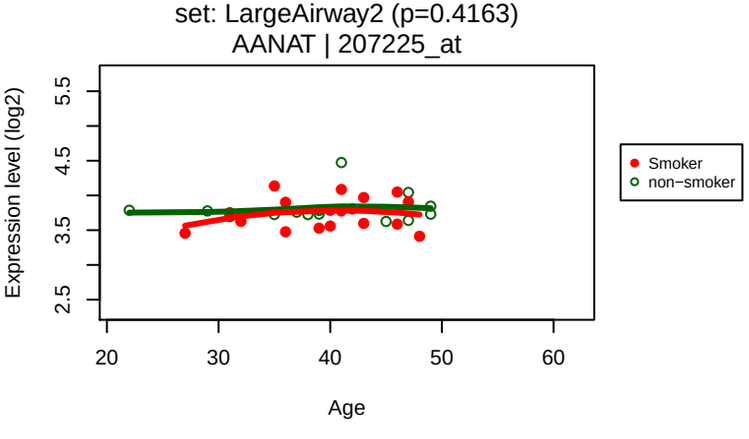

Gene expression ratio (smokers/nonsmokers)
gene: AANAT | 207225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4836
p=0.9506
p=0.7274
p=0.4503
p=0.4163
p=0.5835
p=0.4132
p=0.0349
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

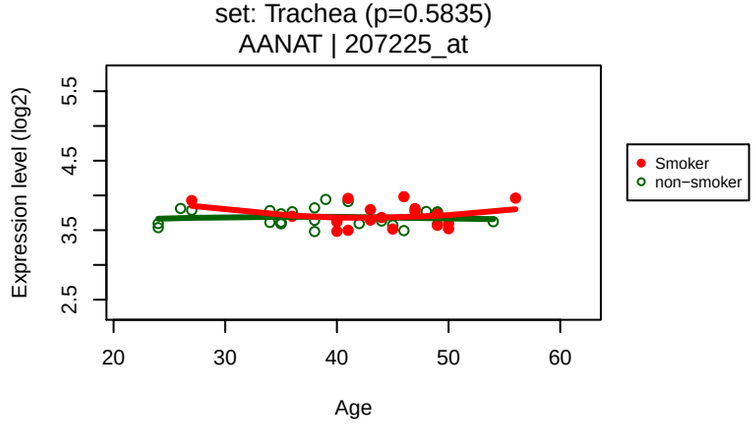

Gene expression ratio in Males
gene: AANAT | 207225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6043
p=0.8979
p=0.2724
p=0.0668
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AANAT | 207225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4975
p=0.1474
p=0.0484
p=0.362
SmallAirway2
LargeAirway2
Trachea
Macrophage2

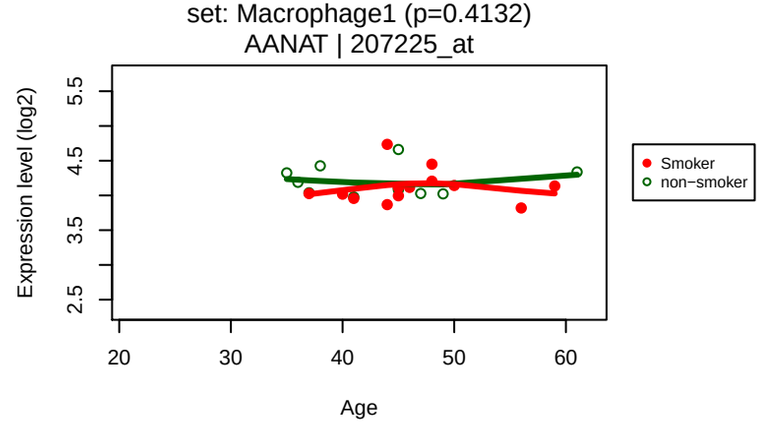

set: SmallAirway1 (p=0.4836)
AANAT | 207225_at
Expression level (log2)
Age
Smoker
non−smoker

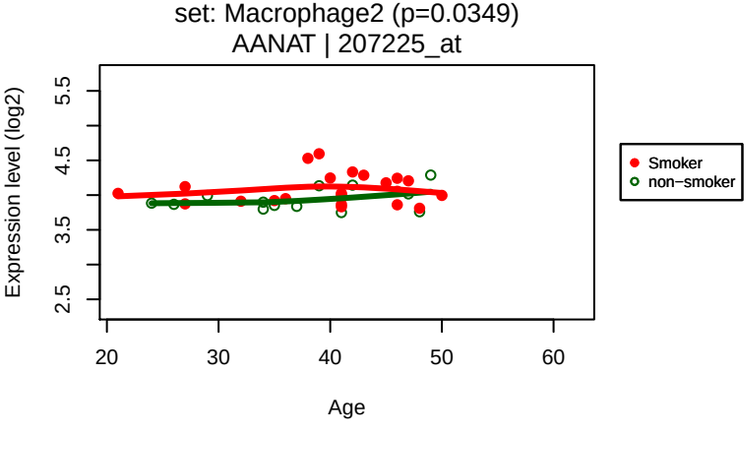

set: SmallAirway2 (p=0.9506)
AANAT | 207225_at
Expression level (log2)
Age
Smoker
non−smoker

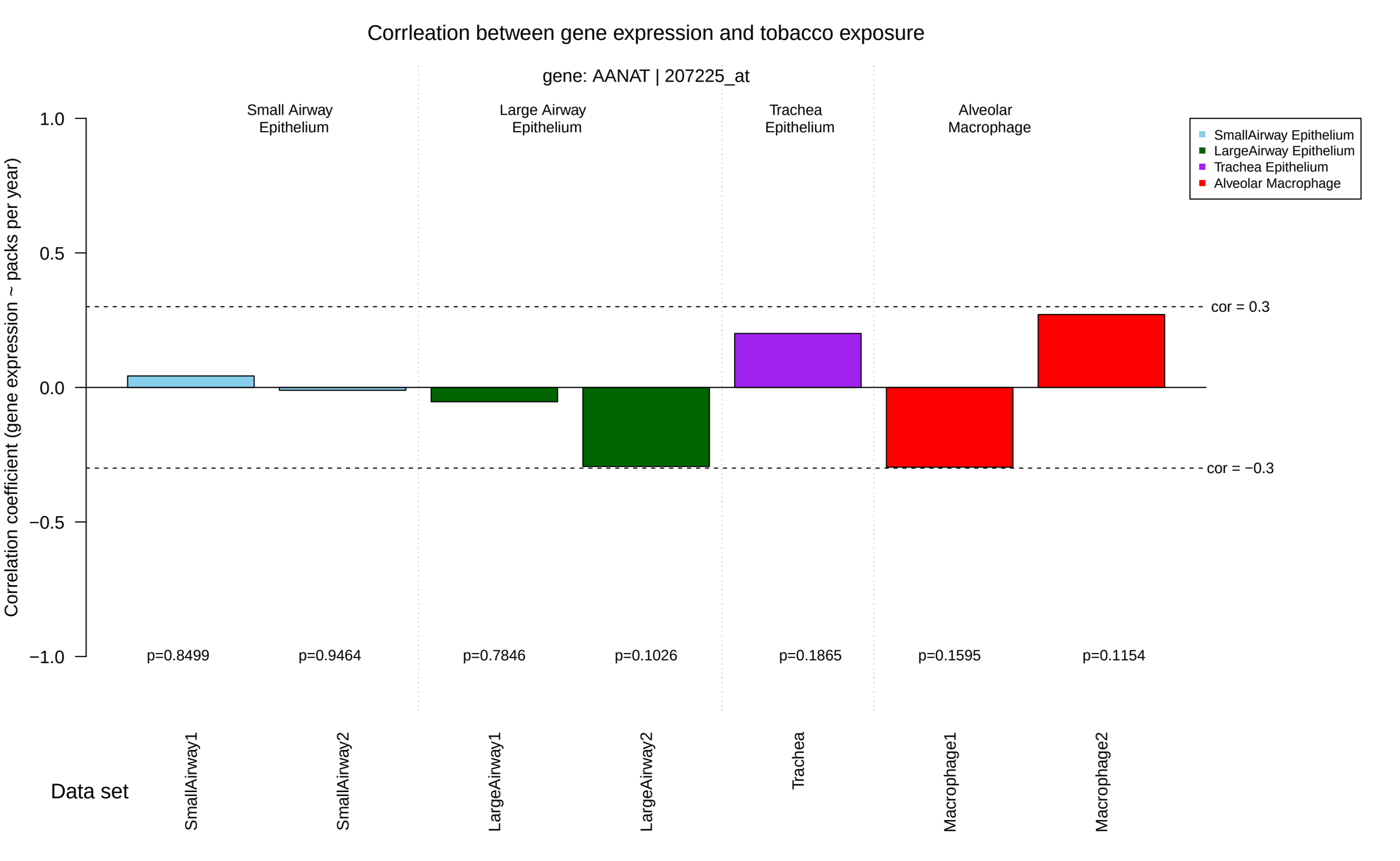

set: LargeAirway1 (p=0.4503)
AANAT | 207225_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4163)
AANAT | 207225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5835)
AANAT | 207225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4132)
AANAT | 207225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0349)
AANAT | 207225_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AANAT | 207225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8499
p=0.9464
p=0.7846
p=0.1026
p=0.1865
p=0.1595
p=0.1154
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2